

1(a)(i)	B prophase ; I early / mid / late C metaphase ; A <u>early</u> anaphase	2
1(a)(ii)	any three from: A identical chromatids for sister chromatids <i>assume points are related to anaphase, unless stated as telophase</i> sister / identical, chromatids (of each chromosome) separate / AW ; A idea of spindle fibres pulling apart sister chromatids <i>in context of end of metaphase / start of anaphase</i> movement of / AW, daughter chromosomes / chromosomes / (sister) chromatids, to (opposite) poles (of cell) ; R if context is a chromosome composed of two chromatids R if context is movement of sister chromatids to same pole I ends of cell movement (to opposite poles) by spindle fibres, contracting / pulling ; I moved by spindle fibres centromeres, qualified ; e.g. centromeres divide at start of anaphase (to separate sister chromatids) centromeres leading (during movement so arms of chromosome lagging) <i>if telophase <u>stated</u>, mark behaviour of chromosomes in telophase to max 2:</i> (daughter) chromosomes at poles ; A sister chromatids at poles A ref. to two separate groups of (daughter) chromosomes (chromosomes) become diffuse / become long and thin / decondense / uncoil ; A become chromatin <i>ref. to re-forming nucleolus ;</i>	3

Question	Answer	Marks
1(a)(iii)	any three from: (active during) DNA replication / S phase / synthesis phase ; A during synthesis of DNA / semi-conservative replication joins / AW, Okazaki fragments ; A description of Okazaki fragments on lagging strand ; (by catalysing) formation of phosphodiester bonds (between adjacent nucleotides) ; to complete sugar phosphate backbone / AW ; e.g. produces continuous, DNA / polynucleotide, strand	3
1(b)(i)	at both ends (of interphase chromosomes) ; A on the ends (of, a chromosome)	1
1(b)(ii)	any three from: telomeres, maintained / AW, for, longer / greater length of time ; allows increased, number of cell cycles / DNA replication cycles / AW ; A allows, mitosis / cell division, to occur more times AW R speeds up, mitosis / cell division / cell cycle <i>idea of</i> prevents loss of genes (at ends of chromosomes, for a longer time) ; A loss of, genetic information / coding DNA A less, genes / AW, lost I loss of genetic material <i>ref.</i> greater ability to replace, damaged / worn out / old, cells ; <i>in context of active for a longer time</i> <i>idea of</i> cell metabolism remains optimal / cell does not go through apoptosis / AW ;	2